Successive h-indices using OpenAlex

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Abstract

I apply Schubert's successive h-index framework to OpenAlex, computing h_2 for 22,980 institutions and h_3 for 214 countries across 13.3 million researchers. The overall institution-level ranking is substantially shaped by biomedical field representation: Medicine accounts for 24% of the author pool and institutions without a large biomedical program are structurally disadvantaged regardless of their non-biomedical strength. Field-specific decomposition is more informative: according to this metric, Wageningen University leads Agricultural & Biological Sciences and the University of Chinese Academy of Sciences leads Environmental Science, Materials Science, and Engineering; Carnegie Mellon leads Computer Science from 159th; Brazil holds two of the world's top-six dental-research institutions by field h_2. I normalize for group size using Egghe's Lotkaian scaling and find that Italy and Japan tie on h_3 despite a 6× difference in institutional scale.

Keywords: successive h-index, OpenAlex, institutional ranking, research assessment

JEL Classification: I23 , C80

MSC Classification: 62P99 , 62-07 , 62P25

1 Introduction

In 2007, András Schubert published a paper titled "Successive h-indices." (2007) It begins:

> Hirsch's excessively discussed h-index generated a long row of interpretations, applications and derivatives within a surprisingly short period. Symptomatically, Hirsch's paper in the *Proceedings of the National Academy*

> *of Sciences of the United States of America* became the second highest cited paper of the issue in which it had been published. In this note, attention is called to an apparently so far overlooked option: h-indices themselves may form the basis of a successive series of h-indices.
>
> Given, e.g., the h-indices of the researchers of an institute (which indices will be denoted by h_1 in what follows), an index, h_2, of the institute can be determined: the institute has an index h_2 if h_2 of its N researchers have an h_1-index of at least h_2 each, and the other $N - h_2$ researchers have h_1-indices lower than h_2 each. The succession can then be continued, e.g., for networks of institutions or countries or other higher levels of aggregation.
>
> Since no sufficient data for testing the concept on the researcher-institute-country hierarchy were available, another example was chosen.

Fifteen years later, OpenAlex (Priem, Piwowar, & Orr, 2022) launched a FOSS bibliographic catalogue complete with citation indexing, authors, fields, and institutions. Recent evaluations have found it to be a broadly suitable substitute for proprietary sources such as Scopus and Web of Science at the institutional level (Alperin, Portenoy, Demes, Larivière, & Haustein, 2024; Culbert et al., 2025), and I take it here as the dataset for which Schubert wished in 2007.

The idea of applying the h-index hierarchically at the researcher–institution level was proposed independently by Prathap (2006) in a one-page note less than a month before Schubert's paper. Schubert's distinctive contribution was to recognize that the construction extends to an arbitrary succession of levels, opening the path to h_3 and beyond. In the years between, successive h-indices attracted modest but sustained attention. Egghe (2008) gave the concept a theoretical footing, showing how the exponent of the underlying Lotka distribution propagates through each successive aggregation step. Several case studies followed shortly after: Ruane and Tol (2008) computed a rational (non-integer) successive h-index across economics departments in Ireland; Arencibia-Jorge, Barrios-Almaguer, Fernández-Hernández, and Carvajal-Espino (2008) applied Schubert's exact researcher–department–institution hierarchy to a Cuban research institute using Web of Science data; and Arencibia-Jorge and Rousseau (2009) applied Prathap's framing to twelve Cuban brain-research institutions. None of this prior work, however, has had access to a comprehensive, open, institution-labeled snapshot of the kind OpenAlex now provides. Yang, Wu, Zhang, Wang, and Deng (2023) subsequently proposed a k-step h-index that extends the *citation graph* at a fixed organizational level: a k=2 h-index for an author counts papers that cite the author's papers. In contrast, Schubert's subscript extends the *organizational hierarchy* from authors to institutions to countries. The two families iterate a common construction along orthogonal dimensions and are complementary rather than competing.

Schubert suggested that h_2 of institutes would be more interesting than that of journals; Section 2 describes how I compute it at scale, and Section 3 presents the results.

2 Methodology

2.1 Data source and filtering

The underlying data is OpenAlex's public authors snapshot (Priem et al., 2022), distributed as Parquet on `s3://openalex/data/parquet/authors/` (`us-east-1`, no credentials required) and partitioned into over 1,600 daily `updated_date=` directories totalling about 53 GB. Only four columns are read from each remote file (`id`, `h_index`, `affiliations`, `topics`), and an author is retained only if all of the following hold:

- `summary_stats.h_index` is non-null and greater than zero;
- at least one entry in `affiliations` has `institution.type = 'education'`;
- at least one entry in `topics` carries a non-null field classification.

Applying these filters to the full snapshot yields 30.0 million authors across OpenAlex's 26 fields. One limitation carried over from the source data: an author's `h_index` is computed by OpenAlex from all works meeting its citation criteria regardless of retraction status, which OpenAlex records inconsistently (Hauschke & Nazarovets, 2025); a small, unquantified fraction of h_1 values in the dataset may therefore be inflated by since-retracted work.

2.2 Author-level assignment

Each retained author is reduced to a single row: one institution and one field. For institution assignment, rather than reading the institution from the author's affiliation list in the snapshot, I derive it from the OpenAlex works snapshot (Parquet files at `s3://openalex/data/parquet/works/`, no credentials required), because the `affiliations[]` field in the author record records only which calendar years an institution appears on the author's papers, not on how many. An authorship in the works snapshot credits education institutions as follows (hereafter called the "rollup" rule, applied by default): every institution with `type = 'education'` listed on the authorship is credited directly; additionally, for every non-education institution listed, its education-type ancestors in OpenAlex's institution `lineage` are credited (so a Jet Propulsion Laboratory authorship credits Caltech); education institutions are never rolled up to their own parents, so a UC Berkeley paper does not also credit the University of California System. An author's primary institution is then the education institution credited on the most of their authorships in the last five publishing years[1] (the author's most recent publication year and the four calendar years before it). Two thresholds must be met for an assignment to be made: the institution must be credited on at least two works in the window, and on at least 10% of the window's works that credit *any* education institution. The second threshold uses the education-crediting works as its denominator (rather than all the author's works) so that hospital- and institute-based faculty whose papers mostly list only their hospital

[1] The five-year window measures each author's primary institutional affiliation during their most recent active publishing period, not current employment. It credits accumulated h_1 to the institution where the author most recently produced research. Inactive or relocated researchers are therefore assigned to their last-active institution rather than their current position, which means the roster measures recent research affiliation rather than a contemporaneous census of current staff.

are not penalised when the hospital has an education-type parent through the rollup rule. Among authors who meet the thresholds, ties go first to the institution seen most recently in the window, then to the institution appearing most often across the author's whole career, then to the numerically smallest OpenAlex institution identifier. Of 126.6 million unique authors appearing in the works snapshot, 41.8 million (33.0%) have at least one education institution credited in their five-year window; 25.9 million fail the minimum-papers threshold, and 2,326 fail the share threshold, leaving 15.9 million with a primary institution assigned. Joining these with the 30.0 million authors in the snapshot who have at least one education affiliation, and at least one topic classification yields the 13.3 million authors across 22,980 institutions that form the basis for all rankings in this paper.

Field assignment works the same way but is weighted. OpenAlex tags each author with a set of topics, each carrying a `count` (roughly, the number of the author's works classified under that topic) and a parent field. I sum `count` within each field and assign the author to whichever field carries the largest total weight: their modal field by publication volume, not simply their single most granular topic. The result is an `authors` table with one row per author: `(author_id, h_index, institution_id, institution_name, field, field_name)`.

Two structural features of this procedure limit the impact of field classification errors. First, the paper operates at the *field* level (26 fields) rather than at the finer topic level (∼4,500 topics); OpenAlex's topic classifier fine-tunes a multilingual BERT model on title, abstract, and citation data and achieves a top-1 accuracy of 0.53 (OpenAlex, 2024), but many topic-level errors stay within the same parent field, so field-level accuracy is plausibly higher, though this is an inference from the structure of the classification hierarchy rather than a directly measured field-level accuracy figure: no field-level validation of the classifier is presented here. Second, an author's field is determined by the mode of their *entire publication record*, not by a single work: a researcher would need a substantial fraction of their career output systematically misclassified to a different field before their assignment would change.

2.3 The h-index and its successive generalization

Hirsch's (2005) original index, here h_1, is computed by OpenAlex per author from citation counts and taken as given. Schubert's generalization (2007) lifts the same definition up a level of aggregation: a group of N members, each already carrying an h_{k-1} index, has an index $\mathrm{h}_k = n$ if n of its members have $\mathrm{h}_{k-1} \geq n$ and the remaining $N - n$ members have $\mathrm{h}_{k-1} < n$:

Algorithm 1 Successive h-index of a group G, given a value $v(m)$ for each member $m \in G$

1: Sort G by v descending; let $r(m)$ be m's 1-indexed rank in that order
2: **return** $\max\{\, r(m) : v(m) \geq r(m),\ m \in G \,\}$

I use this rank-based procedure, implemented as a window function in SQL ranking rows within each group and taking the largest rank whose value has not yet fallen

below it, for every successive index in this paper: h_2 of an institution over the h_1 of its authors, and h_3 of a country over the h_2 of its institutions. The algorithm returns 0 for an empty group or a group in which no member satisfies $v(m) \geq 1$. The SQL implementation uses `ROW_NUMBER() OVER (PARTITION BY group ORDER BY h_index DESC, author_id ASC)`, which assigns sequential integers with ties in `h_index` broken deterministically by `author_id`; this tie-breaking is applied identically in the baseline and in all sensitivity variants. Note that the prose formulation in Schubert's (2007) original description, "the other $N - h_2$ researchers have h_1-indices lower than h_2 each," slightly overstates the condition; the correct definition requires only that *at least* n members have $h_{k-1} \geq n$, with no constraint on the remaining members. For five members with sorted values $(v(m_1), \ldots, v(m_5)) = (5, 4, 4, 4, 4)$, the h-index is 4: m_4 satisfies $v(m_4) = 4 \geq r(m_4) = 4$, but m_5 does not because $v(m_5) = 4 < r(m_5) = 5$.

2.4 h_2: institution and institution×field

h_2 is computed twice, over different groupings of the same author-level data. The field-specific table groups authors by (institution, field) and yields 229,309 pairs; field-specific h_2 uses each author's overall h_1, derived from their full-career publication record across all fields, rather than a citation count restricted to within-field works; it therefore measures the concentration of researchers whose modal field is the assigned discipline and who have strong overall h_1 values, with possible influence from work in other fields. Institutions are matched on their numeric identifier only; institution and field *names* are resolved afterward via `arg_max`, so that spelling or formatting variants of the same name across snapshot partitions cannot silently split one institution into several groups. The institution-overall table ignores field entirely and applies the same procedure across all of an institution's authors regardless of discipline, yielding one h_2 per institution (22,980 rows with a named institution).

2.5 h_3: country level

Institutions are mapped to countries using the `country_code` field of OpenAlex's institutions snapshot (also read directly from S3, joined on institution ID). h_3 is then computed with the same rank-based procedure, once globally per country over the h_2 values of its institutions and once per (country, field) over the field-specific h_2 values, producing one file per field.

2.6 Size-adjusted index

Mryglod, Holovatch, and Kenna (2022) showed that group-level h-indices grow with group size even when per-member quality is held constant. They were not studying successive h indices, but a similar argument makes raw h_2 and h_3 values unsuitable for comparing institutions or countries of unequal scale.

Egghe (2008) applied a Lotkaian framework to successive h indices. Suppose the article–citation frequency function follows Lotka's law with exponent $\alpha_0 > 1$, and the author–article frequency function follows Lotka's law with exponent $\alpha_1 > 1$. Egghe showed that the distribution of h_1 values across authors is itself Lotkaian with compound exponent $\beta_1 = \alpha_0 \alpha_1$. Applying the h-index formula for Lotkaian systems, the

institution-level index satisfies

$$h_2 = S^{1/\beta_1} \tag{1}$$

where S is the number of authors at the institution. An analogous argument at the next level gives the distribution of h_2 values across institutions a Lotkaian exponent $\beta_2 = \alpha_0\alpha_1\alpha_2$, where α_2 is the Lotka exponent of the institution–author frequency function, so that

$$h_3 = R^{1/\beta_2} \tag{2}$$

where R is the number of institutions in the country. Cross-country comparisons of ε_3 should be read relative to OpenAlex's institutional taxonomy: how national research systems are partitioned into distinct institutions (e.g., whether affiliated institutes are credited separately or through a parent institution) affects both h_3 and the institution-count denominator without changing the underlying research activity.

I define size-adjusted metrics by dividing out the full empirically fitted scalings:

$$\varepsilon_2 = \frac{h_2}{c_2\, S^{1/\beta_1}} \tag{3}$$

$$\varepsilon_3 = \frac{h_3}{c_3\, R^{1/\beta_2}} \tag{4}$$

where c_2 and c_3 are the fitted slope coefficients from the no-intercept regressions of h_2 on S^{1/β_1} and h_3 on R^{1/β_2}, respectively (Section 3.9). This normalization ensures that the average institution has $\varepsilon_2 \approx 1$, so $\varepsilon_2 > 1$ means the institution lies above the empirical regression line for its headcount and $\varepsilon_2 < 1$ means it lies below. Ranking by ε_2 is identical to ranking by $h_2/S^{1/\beta_1}$ since c_2 is a constant.

The exponents β_1 and β_2 are estimated directly from the data rather than assembled from separately fit $\alpha_0, \alpha_1, \alpha_2$. I apply discrete maximum-likelihood tail estimation following Clauset, Shalizi, and Newman (2009), with $x_{\min}$ chosen to minimize the Kolmogorov–Smirnov distance between the empirical and fitted CDFs, to the distributions of h_1 across all authors and h_2 across all institutions, reading β_1 and β_2 as their fitted Lotka exponents. The `powerlaw` package parameterizes the discrete power law as $P(X = x) \propto x^{-\alpha}$ for $x \geq x_{\min}$, following Clauset et al.'s survival-function convention; this matches Egghe's Lotka-law density directly, so the fitted α values plug into Egghe's formulas without offset. The standard error of the MLE exponent is approximately $(\alpha - 1)/\sqrt{n_{\text{tail}}}$ (Clauset et al., 2009). I then fit α_0, α_1, and α_2 individually: α_0 from a stratified sample of per-article citation counts drawn from the OpenAlex works snapshot, α_1 from per-author work counts, and α_2 from per-institution author counts. Finally, I verify that $\alpha_0\alpha_1 \approx \beta_1$ and $\alpha_0\alpha_1\alpha_2 \approx \beta_2$ as a cross-check that the compounding-exponent structure of Egghe's model holds in the OpenAlex dataset.

2.7 Supplementary metrics

Several secondary statistics are derived from the h_2 and h_3 tables to probe what the headline numbers obscure:

- **Gini coefficient**, computed on the within-field distribution of institutional h_2, quantifies how concentrated institutional h_2 is within a field (Gini near 1) or how broadly distributed (Gini near 0).

- **Breadth score** counts, per institution, how many of the 26 fields it ranks in the global top 10, top 50, and top 100.
- **Non-biomedical ranking** recomputes institution-level h_2 after excluding authors whose modal field is Medicine, Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, or Health Professions, to test whether the overall ranking is, in substance, a ranking of medical school size.

2.8 Manual cross-validation

Because errors in OpenAlex's underlying author disambiguation or citation counts would propagate directly into every successive index, I manually spot-check a targeted, non-random sample of authors rather than a random one, chosen at the points where an error would be both most likely and most consequential: authors sitting exactly at an institution's h_2 threshold, where a ± 1 error in a single h_1 could change the institution's rank; top authors in fields where OpenAlex's topic classification is known to be thinner (e.g., Arts and Humanities); top authors at institutions with non-Anglophone naming conventions, where author-name disambiguation is more error-prone and where OpenAlex's coverage is known to diverge unevenly from other databases by country, most severely for China (Zheng, Miao, Bu, & Larivière, 2025), including institutions suspected of affiliation-swapping for grant purposes (Bhattacharjee, 2011); and the small number of authors with the highest h_1 globally, as a sanity check at the extreme of the distribution. For each candidate, I resolve the author's OpenAlex profile through the public API and compare it by hand against Web of Science, Scopus, and Google Scholar. This design is deliberately adversarial: it is built to surface errors, not to estimate how frequently they occur across the full population, so any outlier rate found in it bounds the qualitative failure mode (which authors, and by how much) rather than the population-wide error rate.

2.9 Sensitivity to disambiguation error

Because the cross-validation sample above cannot itself bound a population-wide error rate, I separately ask how much author-disambiguation error the paper's conclusions could tolerate before changing using a Monte Carlo sensitivity analysis run directly on the full authors table. At a population-wide error rate p, I flag each author independently with probability p. For authors affiliated with institutions in China, South Korea, Iran, or Saudi Arabia, I flag each author with probability $\min(1, 3p)$, reflecting the cross-validation finding that disambiguation errors concentrate in non-Anglophone, common-name strata. A flagged author's h_1 is perturbed by a ratio drawn uniformly from the four ratios observed among the cross-validation outliers (103/57, 45/11, 215/115, 139/41), with equal probability of upward inflation ($\times$ ratio, simulating a merged-identity error) or downward deflation ($\times$ ratio^{-1}, simulating a split-identity error). I recompute institution-level h_2, country-level h_3, and the field-specific h_2 rankings, and compare each to the unperturbed baseline via Spearman rank correlation and top-set overlap. Rather than picking a single rate to call a "worst case," I run fifty independent trials at each of seven rates, $p \in \{1\%, 5\%, 10\%, 13\%, 20\%, 40\%, 80\%\}$, tracing out how ranking stability responds

to the assumed amount of error from a plausible rate up to a deliberately implausible one. In all scenarios the perturbation holds institutional roster membership fixed; actual merges and splits can also move authors between institutions, which is not modelled here and is acknowledged as a limitation of this approach.

Three additional scenarios probe the sensitivity of the baseline results to modelling assumptions. (1) *Directional inflation-only*: all flagged authors receive upward perturbation only (simulating a pure over-merge environment with no corrective splits). (2) *Threshold-concentrated*: only authors with h_1 within ± 5 of their institution's baseline h_2 are eligible for perturbation, with perturbation bidirectional for those flagged; this targets the authors who most directly determine rank-threshold crossings. (3) *Uniform (non-stratified)*: the bidirectional model is run without the $3\times$ country multiplier, applying the same base rate p uniformly to all authors, to test how much of the baseline scenario's instability is attributable to the elevated rate for CN/KR/IR/SA institutions.

2.10 Robustness to field assignment

The single-field assignment described in Subsection 2.2 asks "which field accounts for the largest share of an author's publication volume?" and assigns each author exclusively to that field. To test whether field-specific h_2 rankings are stable under an alternative that allows multidisciplinary authors to count in multiple fields simultaneously, I re-run the field-specific h_2 computation after expanding each author's field membership: at a threshold τ, an author is assigned to every field in which their OpenAlex topic-weight share is at least τ (where topic-weight is the same count-weighted field sum used in the original assignment). I test two thresholds, $\tau \in \{20\%, 33\%\}$, spanning from liberal (an author need contribute only one fifth of their output to count in a field) to moderate (one third or more). For each threshold I recompute h_2 for every (institution, field) pair in the expanded author roster, using the same rank-based algorithm as the original, and compare to the baseline via Spearman rank correlation on shared (institution, field) pairs and by checking whether each field's leading institution remains #1.

2.11 Robustness to institution assignment

Two classes of robustness test bear on the primary-institution assignment. The first asks whether the single-institution constraint itself distorts results. The primary rule credits each author to the one institution with the most works in their five-year window; a *full-counting variant* instead credits each author's full h_1 to every institution meeting both thresholds in that window. Fractional counting would require splitting the h_1 contribution across institutions, which is not straightforward for an integer-valued rank-threshold index; a weighted threshold construction would be needed and is beyond the scope of this paper. This variant directly tests whether multiply-affiliated researchers distort the single-institution rankings.

The second class asks how sensitive the results are to the three independently variable choices *within* the works-based rule: (1) whether to roll up non-education institutions to their education-type ancestors in the OpenAlex lineage (the "rollup" rule);

(2) whether the 10% institution-share threshold uses education-crediting works or all of the author's window works as its denominator; and (3) whether to use the works-based assignment at all, versus reading the most recent education institution directly from the author snapshot. I test each choice independently, producing a *no-rollup* variant that suppresses ancestor credit (so a Jet Propulsion Laboratory authorship no longer credits Caltech); a *share-of-all* variant that computes the share over all window works rather than only education-crediting ones; and a *most-recent-snapshot* variant that assigns each author to the education institution appearing most recently in the snapshot, breaking ties by the most frequent career-wide appearance and then by smallest OpenAlex identifier. For all four variants, I recompute institution-level h_2 and country-level h_3 and compare against the paper's default assignment via Spearman rank correlation and top-10 overlap.

2.12 External validation against Scimago Institutions Rankings

To assess whether ε_2 captures genuine research quality rather than a residual size artefact, I compare raw h_2 and ε_2 against the Scimago Institutions Rankings (SIR) 2026 "Overall Rank" export (SCImago Research Group, 2026), a Scopus-derived composite ranking (Research 50%, Innovation 30%, Societal 20% sub-scores). I compute Spearman ρ between SIR's Global Rank and both the raw h_2 rank and the ε_2 size-adjusted rank and repeat the comparison restricted to SIR's "Universities" sector only, excluding the ministries, hospitals, and companies that SIR also ranks, to check whether any mismatch is an artefact of comparing against a mixed-sector list.

2.13 Software and reproducibility

All analyses were implemented in Python 3.10 using NumPy (Harris et al., 2020), pandas (Wes McKinney, 2010), geopandas (Fleischmann et al., 2026), matplotlib (Hunter, 2007), duckdb (Mühleisen & Raasveldt, 2026), powerlaw (Alstott, Bullmore, & Plenz, 2014), and scipy (Virtanen et al., 2020). The data fetching, filtering, and processing are fully reproducible from the public code repository at https://github.com/rupumped/successive-h-indices. The exact OpenAlex snapshot used here consists of the authors and works Parquet partitions through `updated_date=2026-06-26`, both from the public S3 bucket `s3://openalex/data/parquet/` (`us-east-1`). All tables in this paper were produced from git commit `e8f0671`. Table 1 summarises the exclusion counts from raw snapshot to final analysis table.

Table 1 Data-flow summary: exclusions and row counts from raw OpenAlex authors snapshot to final analysis table.

Step	Authors retained	Removed
Raw authors snapshot (through 2026-06-26)	100M	—
Has at least one affiliation record	30M	70M
Works-based primary institution assigned	15.9M	14.1M
$h_1 > 0$ filter	13,311,757	2,588,243

3 Results

3.1 Manual cross-validation

Of the 30 authors sampled under the scheme described in Subsection 2.8, Table 2 summarizes agreement between OpenAlex's `h_index` and the corresponding value I located by hand in Web of Science, Scopus, and Google Scholar. Coverage itself varies sharply: Scopus returned a value for 26 of the 30 authors, Google Scholar for 19, and Web of Science for 16.

Table 2 Agreement between OpenAlex and each external source, over authors for which that source returned a value. Diff is OpenAlex minus the external value.

Source	n	Mean diff	Mean abs. diff	Worst case
Web of Science	16/30	+24.5	25.7	Lei Jiang (+100)
Scopus	26/30	+8.0	12.4	Donald Small (+46)
Google Scholar	19/30	−22.7	30.7	A. Jafari (−85)

Scopus tracks OpenAlex most closely, both in coverage and in magnitude of disagreement. Google Scholar is systematically higher than OpenAlex, consistent with its broader crawl of preprints, theses, and other grey literature and its comparative permissiveness toward self-citation; this ordering (Google Scholar > Scopus > OpenAlex > Web of Science) reproduces cleanly among the six global top-h_1 sanity-check authors, none of whom show a discrepancy exceeding 27% of their OpenAlex value in either direction.

Four authors, however, show disagreement well outside this range. I read this as a signature of author-disambiguation failure (multiple distinct people merged under one OpenAlex author ID) rather than of genuine cross-database disagreement:

- **Donald Small** (Johns Hopkins): $h_1^{\text{OpenAlex}} = 103$ against Scopus = 57 and Google Scholar = 59.
- **Karen S. Calhoun** (University of Michigan): $h_1^{\text{OpenAlex}} = 45$ against Scopus = 11 and Google Scholar = 13.
- **Lei Jiang** (University of Science and Technology of China): $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115, consistent with documented China-specific coverage discrepancies in OpenAlex (Zheng et al., 2025).
- **A. Jafari** (Sharif University of Technology): $h_1^{\text{OpenAlex}} = 139$, with Scopus = 133 and Google Scholar = 224; Web of Science reports only 41.

The first three cases show large OpenAlex–Scopus discrepancies (factors of 1.8–4.1×) consistent with multi-person merges. The A. Jafari case is different: OpenAlex (139) and Scopus (133) agree closely (4% difference), and the low Web of Science value (41) is more plausibly explained by WoS's documented undercoverage of non-Anglophone institutions than by an OpenAlex over-merge. Cross-database h-index differences alone cannot distinguish disambiguation errors from coverage differences; publication-level

identity verification (matching individual works across databases) would be needed for a definitive conclusion and is beyond the scope of this paper. We retain A. Jafari in the perturbation ratio pool as a conservative choice but note that this case is less clearly a disambiguation failure than the other three. The three more definitive cases fall in the non-Anglophone-institution or common-name strata of the sample, matching the concern raised in Subsection 2.8 that disambiguation is more error-prone for exactly these authors. Because such merges can only inflate an author's apparent h_1, and h_1 feeds directly into h_2 and h_3, a handful of merged identities at the right institution could measurably distort an institution- or country-level ranking; I flag this as an open limitation rather than one this sample can size definitively. This sample is deliberately adversarial, chosen to find errors, not to estimate how often they occur. The 4/30 (13%) outlier fraction should not be read as an estimate of how much of the full dataset is affected. The same pattern, disambiguation precision concentrated lower for common names and non-Anglophone institutions, appears in an independent benchmark of 2,700 scholars sampled systematically across three regions (Zhao & Chen, 2025), corroborating that this is a geographically and onomastically concentrated failure mode rather than an artifact of this particular sample. OpenAlex's maintainers report an ongoing correction effort targeting exactly this kind of over-merging, with corresponding-author identification precision rising from 0.60 to 0.92 (F1 0.72 to 0.90) after a 2026 rework of the underlying author-matching model (Demes, 2026). Subsection 3.12 quantifies how much a population-wide error rate at or below the adversarial sample's raw rate would actually move this paper's rankings.

The four Arts and Humanities threshold authors sampled (Gillian Ramchand and G. Herbert Fowler at Oxford; D.H. Mellor and James Raven at Cambridge) illustrate the range of under-coverage across major citation databases: Ramchand and Fowler returned no value in any of the three external sources, while Mellor (OA $h_1 = 28$) and Raven (OA $h_1 = 24$) appeared in Scopus at roughly half their OpenAlex values ($h_1 = 13$ and 12, respectively); for Raven, Google Scholar (26) brackets the OpenAlex value closely. This is consistent with the well-documented weaker coverage of book-based humanities scholarship in all major citation databases (Nederhof, 2006) and suggests that h-index comparisons in these fields should be read with correspondingly less confidence than in the sciences.

3.2 The overall h_2 ranking

Table 3 lists the ten institutions with the highest institution-level h_2, computed over all 22,980 institutions and 13.3 million authors described in Section 2.

This broadly tracks the US News and World Report ranking, where Harvard also places 1st, Stanford 3rd, and Cambridge 5th. As I show below, however, the overall ranking is a considerably less interesting object than it first appears.

3.3 Field-level heterogeneity

Splitting h_2 by (institution, field) reveals that the overall ranking is deceptive: several institutions that never come close to the top of Table 3 rank first globally in their field by this measure. Table 4 lists a sample of these cases. A note on interpretation:

Table 3 Top 10 institutions by overall h_2.

Institution	h_2	Authors
Harvard University	135	47,279
Stanford University	113	36,976
University of Antwerp	106	7,207
University of California San Diego	106	24,953
University of Cambridge	106	24,810
University of Oxford	105	28,577
University of Washington	104	30,392
Johns Hopkins University	102	33,329
Massachusetts Institute of Technology	101	21,527
University College London	101	33,413

field-specific h_2 uses each author's overall h_1 rather than a citation count restricted to within-field works. It measures the concentration of researchers whose modal field is the assigned discipline and who have strong overall h_1 values.

University of Antwerp is the counterexample that proves the rule holds both ways: it is #3 overall ($h_2 = 106$) *and* #1 in Physics and Astronomy, a broad powerhouse that also ranks first in a specific, non-biomedical field. Universidade Estadual de Campinas (UNICAMP) places 6th in Dentistry behind São Paulo, placing Brazil among the top six globally in Dentistry by this measure despite the country appearing nowhere near the top of any general-purpose ranking (Vaccaro, Sánchez-Núñez, & Witt-Rodríguez, 2022). China's leading field-level h_2 positions in applied sciences are concentrated in the University of Chinese Academy of Sciences (UCAS, rank 24 overall, h_2=95), which leads Environmental Science, Materials Science, Engineering, and Chemical Engineering by field h_2, while its sister institution, the University of Science and Technology of China (USTC, rank 66, h_2=84), leads Energy.

Not every field supports this kind of comparison equally well. Decision Sciences is the thinnest field in the dataset: Stanford's global #1 is $h_2 = 20$, built on only 129 Stanford authors against 56,987 authors field-wide, versus Medicine's $h_2 = 118$ built on 3.23 million authors field-wide. h_2 is not meaningfully comparable across fields of such different size, only within a field, and Decision Sciences in particular may be too sparse an OpenAlex topic category for its ranking to carry much signal.

3.4 Robustness of field-specific rankings to multi-field assignment

The single-field assignment described in Subsection 2.2 asks which field accounts for the plurality of an author's publication volume, and assigns them exclusively there. To test how much this choice moves the field-level rankings, I re-run the field-specific h_2 computation after allowing each author to count in every field for which their OpenAlex topic-weight share is at least τ in accordance with Subsection 2.10.

At $\tau = 33\%$ (an author must contribute at least one third of their output to count in a field), the (institution, field) pair count rises by 16.0% (229,309 $\rightarrow$ 266,002 pairs). The Spearman rank correlation between original and multi-field h_2 values, computed

Table 4 Selected field leaders that rank well outside the overall top 10. "Overall" columns give the institution's rank and h_2 in Table 3's full ranking; "Field h_2" is the institution's h_2 restricted to that field.

Field	Leader	Ov. rank	Ov. h_2	Field h_2
Agricultural & Biological Sciences	Wageningen	148	73	58
Environmental Science	Univ. of Chinese Academy of Sciences	24	95	60
Pharmacology, Toxicology & Pharmaceutics	Shenyang Pharmaceutical University	957	40	26
Computer Science	Carnegie Mellon University	159	71	56
Materials Science	Univ. of Chinese Academy of Sciences	24	95	71
Engineering	Univ. of Chinese Academy of Sciences	24	95	71
Energy	USTC	66	84	51
Chemical Engineering	Univ. of Chinese Academy of Sciences	24	95	30
Dentistry	Universidade de São Paulo	226	66	37

on the inner join of shared pairs, is $\rho = 0.960$: field rankings are highly stable. Of the 28 fields' leading institutions (26 fields, but Social Sciences and Business, Management and Accounting each have two institutions tied for the top h_2 value, yielding 28 leader entries), 23 are unchanged. The Wageningen Ag&Bio leadership (Table 4) survives, Carnegie Mellon's Computer Science lead survives (tied with MIT, $h_2 = 56$), and the University of Chinese Academy of Sciences (UCAS) Environmental Science lead ($h_2 = 62$, baseline $h_2 = 60$) also survives. The two methods answer different questions. The single-field assignment identifies institutions where Environmental Science is researchers' *primary* field; the multi-field assignment identifies institutions with the most researchers for whom it is a *substantial* field.

At $\tau = 20\%$ (a more aggressive threshold admitting all authors with one fifth or more of their output in a field), the pair count rises by 39.6% (229,309 to 320,067) and $\rho = 0.901$; 21 of 28 field leaders survive. Carnegie Mellon's Computer Science lead also falls at this threshold, with Stanford leading instead ($h_2 = 62$), consistent with the same mechanism of Stanford having a larger population of researchers with at least 20% CS output, even if CS is not their modal field. These findings confirm that the

overall field ranking structure is broadly stable under multi-field assignment but that the field-leadership claims for smaller, highly specialised institutions depend on how narrowly "counts in this field" is defined.

3.5 Robustness of institutional rankings to attribution method

Table 5 reports rank-correlation statistics for the four sensitivity variants described in Subsection 2.11, each compared against the paper's default primary-institution assignment. Institution-level h_2 Spearman rank correlations ρ are computed over institutions shared between the default and variant rosters; country-level h_3 correlations use the fixed institution-to-country mapping.

Table 5 Sensitivity of institution h_2 and country h_3 rankings to four alternative attribution methods, each compared against the default primary-institution assignment. Spearman ρ is computed over shared institutions (for h_2) or shared countries (for h_3); top-10 overlap counts institutions/countries common to both top-10 lists.

Attribution variant	h_2 ρ	h_2 top-10	h_3 ρ	h_3 top-10
Full-counting	0.9391	7/10	0.9954	9/10
No rollup	0.9985	9/10	0.9976	10/10
Share of all works	0.9986	9/10	0.9973	10/10
Most-recent snapshot	0.9074	7/10	0.9906	9/10

The no-rollup and share-of-all variants are nearly indistinguishable from the default: institution h_2 Spearman correlations of 0.998–0.999 and top-10 overlaps of 9/10 confirm that neither the rollup rule nor the denominator choice for the 10% share threshold materially shifts the rankings.

The most-recent-snapshot variant, which reverts to reading each author's primary institution directly from the OpenAlex author snapshot rather than from the works-based five-year window, produces a lower institution h_2 Spearman correlation of 0.907. Seven of the ten top-ranked institutions are shared with the default top-10, but the three that differ include institutions whose apparent performance under the most-recent-snapshot rule reflects single-paper secondary affiliations rather than sustained research presence, the pattern the primary-institution rule was designed to exclude. Country h_3 is more stable ($\rho = 0.991$, 9/10 top-10 overlap), consistent with country aggregation smoothing institution-level noise. These results confirm that the choice between snapshot-based and works-based attribution is the dominant source of sensitivity in this paper's rankings, while choices within the works-based approach (rollup, denominator) are comparatively minor.

Rather than assigning each author to a single institution, the full-counting variant credits each author's full h_1 to every institution meeting both thresholds in their five-year window. Of the 15.9 million authors with a works-based primary institution assigned, 18.0% (2,858,984) qualify at two or more institutions under this rule, producing 17.2 million (author, institution) attributions. Institution h_2 Spearman correlation

is $\rho = 0.939$ (7/10 top-10 overlap); country h_3 is $\rho = 0.995$ (9/10 top-10 overlap). The three institutions entering the full-counting top-10 that were not in the default top-10 (the University of Adelaide, Université Paris-Saclay, and Université Paris Cité) are large research universities whose researchers commonly hold dual appointments with national research agencies (CNRS, 2026; Inserm, 2021), so they gain more authors under a full-counting rule. The overall structure of the rankings is broadly stable at both the institution and country level.

3.6 Concentration within fields

Table 6 reports, for every field, the Gini coefficient of institutional h_2 defined in Section 2.7, together with the field's institution count, leading (maximum) h_2, and mean h_2.

Table 6 Gini coefficient of institutional h_2 within each field, sorted from most to least concentrated.

Field	Institutions	Max h_2	Mean h_2	Gini
Physics and Astronomy	9,054	104	6.42	0.619
Biochemistry, Genetics & Molecular Biology	11,181	100	6.88	0.590
Neuroscience	7,808	62	4.94	0.581
Medicine	16,228	118	7.40	0.579
Earth and Planetary Sciences	7,787	58	4.74	0.573
Immunology and Microbiology	6,095	63	4.46	0.568
Materials Science	10,008	71	6.75	0.562
Environmental Science	12,793	60	5.58	0.557
Engineering	14,486	71	6.78	0.556
Agricultural and Biological Sciences	12,877	58	5.45	0.535
Chemistry	8,463	50	5.35	0.535
Psychology	11,486	56	3.83	0.530
Dentistry	4,712	37	3.65	0.527
Computer Science	14,392	56	5.06	0.519
Energy	6,246	51	3.92	0.514
Mathematics	8,046	35	3.74	0.490
Social Sciences	17,646	46	4.15	0.488
Economics, Econometrics and Finance	10,238	54	3.47	0.486
Business, Management and Accounting	11,909	32	3.77	0.485
Health Professions	10,317	40	3.05	0.479
Veterinary	2,162	26	2.52	0.468
Pharmacology, Toxicology and Pharmaceutics	4,392	26	2.96	0.456
Arts and Humanities	10,759	28	2.70	0.451
Chemical Engineering	3,486	30	2.88	0.443
Nursing	5,333	32	2.63	0.435
Decision Sciences	6,961	20	2.63	0.432

Physics and Astronomy is the most unequal field: University of Antwerp's $h_2 = 104$ is 16.2 times the field mean of 6.4. Decision Sciences is the most evenly distributed (Gini ≈ 0.43). The Gini coefficient here measures inequality in the distribution of h_2 scores across institutions within a field, not inequality in research output or citation impact. Institutions with no authors assigned to a field are excluded from that field's Gini computation; their inclusion as $h_2 = 0$ would increase all Gini values.

3.7 The overall ranking as an implicit biomedical ranking

Medicine alone accounts for 3.23 million of the dataset's 13.3 million authors (24%), and, together with Biochemistry/Genetics/Molecular Biology, Immunology and Microbiology, Neuroscience, and Health Professions, the five core biomedical fields dwarf every other field's author pool. Table 7 recomputes institution-level h_2 after excluding authors in these five fields entirely to test how substantially the overall ranking is shaped by biomedical field representation.

Table 7 Top 10 institutions by non-biomedical h_2, alongside their position in the unrestricted overall ranking. Δrank is overall rank minus non-biomedical rank; positive means the institution moves up once biomedical fields are excluded.

Institution	Rank	h_2	Overall rank	Overall h_2	Δrank
University of Antwerp	1	105	3	106	+2
Harvard University	2	100	1	135	−1
Stanford University	3	100	2	113	−1
Université Paris-Saclay	4	95	14	100	+10
University of California, Berkeley	5	94	16	99	+11
Univ. of Chinese Academy of Sciences	6	93	24	95	+18
Massachusetts Institute of Technology	7	91	9	101	+2
California Institute of Technology	8	90	26	93	+17
ETH Zurich	9	88	35	90	+26
The University of Adelaide	10	88	34	91	+23

The University of Antwerp takes first place without biomedical authors (non-bio h_2=105, overall rank 3), followed by Harvard and Stanford (both non-bio h_2=100, overall ranks 1 and 2 respectively). Antwerp's top position reflects high h indices across physics, chemistry, and life-sciences disciplines outside the core biomedical fields. The table reshuffles substantially: Paris-Saclay, Berkeley, UCAS, Caltech, and ETH Zurich all rise 10–26 places once Medicine and its four sibling fields are set aside, while Cambridge falls 6 places and Oxford 8. The University of Chinese Academy of Sciences is 6th non-biomedical (overall 24th), and The University of Adelaide enters the top 10 (10th non-bio, 34th overall). Because medical fields have by far the largest, most-cited author pools, they structurally dominate any university-wide aggregate; institutions without a large biomedical program, such as USTC or Carnegie Mellon, are disadvantaged in the overall ranking regardless of their non-biomedical h_2. Antwerp's simultaneous top-3 overall and top-1 non-biomedical position shows that a non-biomedical institution can achieve a high h_2 if its non-biomedical h_2 is sufficiently

concentrated; the overall ranking is substantially shaped by biomedical representation, though not reducible to it. The field-level breakdowns matter more than the unrestricted university-level number.

3.8 Breadth versus depth

Table 4 showed institutions that lead a single field while sitting far outside the overall top 100. Table 8 shows the opposite pattern: institutions whose strength is broad rather than deep, measured by how many of the 26 fields they place in the global top 50.

Table 8 The nine institutions with the most top-50 field placements out of 26 fields; ties broken by top-10 field placements.

Institution	Top 50	Top 10	Overall rank	Overall h_2
University of Oxford	19	5	6	105
Harvard University	18	12	1	135
University of Michigan	18	7	22	96
Stanford University	17	13	2	113
University of Toronto	17	5	13	100
University of California, Berkeley	16	4	16	99
University of Cambridge	16	3	5	106
The University of Melbourne	16	1	31	92
University College London	15	8	10	101

Oxford leads on top-50 h_2 breadth (19 of 26 fields), while Stanford leads on top-10 field placements with 13 of 26 fields, narrowly ahead of Harvard's 12. Five of the nine broadest institutions (Oxford, Toronto, Berkeley, Cambridge, and Melbourne) place fewer than half their top-50 fields in the top 10, meaning most of their breadth reflects consistent competence across many disciplines rather than dominance of any one. The University of Melbourne (rank 31, $h_2 = 92$) is the only institution outside North America and the United Kingdom in this group. The contrast between breadth and depth is illustrated by comparing University College London ($h_2 = 101$, 15 top-50 fields) with Oxford ($h_2 = 105$, 19 top-50 fields): UCL places in the top 10 in 8 fields versus Oxford's 5, despite placing in fewer top-50 fields overall. At the other extreme, Imperial College London (rank 21, $h_2 = 96$) achieves its overall position with zero top-10 field placements across 8 top-50 fields, through consistent mid-table performance rather than any single-field depth.

3.9 Institution size-adjusted index

Model validation

Figure 1 plots institution-level h_2 against the Egghe-predicted value S^{1/β_1} for all institutions with at least 10 OpenAlex authors. A no-intercept regression, appropriate because $N = 0$ authors forces $h_2 = 0$, gives slope $c_2 = 2.422$ and uncentered $R^2 = 0.909$ (centered $R^2 = 0.810$): the Egghe functional form $h_2 \propto S^{1/\beta_1}$ accounts

for the large majority of variation in h_2 across institutions. The size-adjusted index $\varepsilon_2 = h_2/(c_2 S^{1/\beta_1})$ captures each institution's deviation from this empirical trend; $\varepsilon_2 > 1$ indicates above-average performance relative to the fitted regression line, and $\varepsilon_2 < 1$ below-average.

As a cross-check, an OLS log-log regression of $\log(h_2)$ on $\log(S)$ gives slope $b = 0.404$ (implying $\beta \approx 2.47$, centered $R^2 = 0.795$), versus Egghe's $\beta_1 = 2.991$ (implied slope 0.334), meaning the Egghe-based normalization slightly under-adjusts for size.

The tail-fit diagnostics (Section 2.6) are: for β_1, $x_{\min} = 18$, $n_{\text{tail}} = 199{,}953$ (2M-point subsample of 13.3M), KS $D = 0.063$, $\text{SE}(\beta_1) \approx 0.004$; a loglikelihood-ratio test comparing power-law vs. exponential within the tail gives $R = -25.3$ ($p \approx 0$), favouring the exponential in the extreme upper tail ($h_1 \geq 18$). For β_2: $x_{\min} = 18$, $n_{\text{tail}} = 3{,}918$, KS $D = 0.064$, $\text{SE}(\beta_2) \approx 0.030$; LRT $R = -9.0$ ($p \approx 0$), again favouring the exponential above $h_2 = 18$. These LRT results indicate that within the extreme upper tail the exponential fits marginally better than a strict power law, while the regression evidence above shows that the power-law functional form for size-scaling is empirically well-supported across the full institution-size range. Institutions with $h_2 < 18$ lie outside the fitted tail domain.

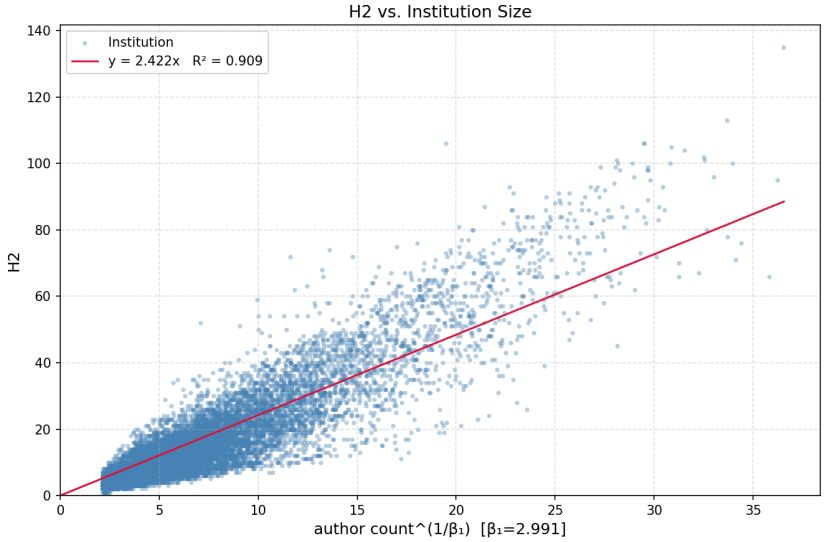


Fig. 1 Institution-level h_2 versus S^{1/β_1} (Egghe's predicted scaling, $\beta_1 = 2.991$). No-intercept regression: $h_2 = 2.422\, S^{1/\beta_1}$, uncentered $R^2 = 0.909$ (centered $R^2 = 0.810$).

The cross-check on Egghe's compounding structure fails sharply, however. Egghe's derivation requires that the Lotka tail exponent of the h_1 distribution equals the product $\alpha_0\alpha_1$; these separate fits give $\alpha_0 = 2.454$ and $\alpha_1 = 2.994$, whose product is 7.35, against the directly-measured $\beta_1 = 2.991$: a discrepancy of 146%. The analogous prediction at the next level ($\beta_2 = \beta_1\alpha_2$) is similarly violated: $2.991 \times 1.987 = 5.944$ against the directly-measured $\beta_2 = 2.847$ (109% discrepancy).

The failure of the compounding formula does not, however, undermine the size-adjusted index. The two claims play different roles. The compounding formula ($\beta_1 = \alpha_0\alpha_1$) is a *theoretical prediction* about the relationship between separately-fitted exponents; it is tested here and found wanting. The power-law scaling $h_2 \propto S^{1/\beta_1}$ is an *empirical description* (uncentered $R^2 = 0.909$, centered $R^2 = 0.810$) that does not depend on the compounding formula being correct. β_1 is estimated directly from the distribution of h_1 values across authors, not derived from $\alpha_0\alpha_1$, so the 146% discrepancy in the compounding formula has no bearing on the exponent actually used in ε_2. The key assumption linking the pooled β_1 to each institution's threshold is that institutional author rosters are approximately random draws from the population-wide h_1 distribution. If institutions instead selectively recruit high-h_1 talent, this assumption is violated; the fitted amplitude $c_2 = 2.422$ and the residual size association documented below are consistent with such quality-sorting, though they do not establish it as the cause.

Size-adjusted rankings

The size-adjusted index $\varepsilon_2 = h_2/(c_2\, S^{1/\beta_1})$ addresses Mryglod et al.'s (2022) concern: the Egghe functional form accounts for the large majority of variation in h_2, and dividing it out isolates the residual variation not explained by headcount alone. However, a regression of $\log(\varepsilon_2)$ on $\log(S)$ across institutions with ≥ 100 authors ($n = 8{,}118$) yields slope $= 0.073$, $R^2 = 0.062$: a small but statistically significant positive residual association with size remains after normalization. Larger institutions still score slightly higher on ε_2 than the regression line predicts. The normalization substantially reduces but does not eliminate the size association; results should be read with this caveat, particularly for comparisons between very small and very large institutions. Field composition and researcher career age are additional uncontrolled confounds. A regression of field-specific ε_2 would be the appropriate next step and is left as future work. Career age is not available in the OpenAlex snapshot columns used here and is likewise acknowledged as a limitation.

Table 9 lists the top 10 institutions by ε_2 among the 8,118 with at least 100 authors. The top of the ranking is dominated by small and highly specialized institutions. Focused research centers and specialized colleges recur throughout the top 20.

The high-ε_2 entries are a mix of specialized research institutes (Center for Astrophysics Harvard & Smithsonian, Stockholm Resilience Centre, Barcelona Institute of Science and Technology), a learned society (Académie Nationale de Médecine), medical and health sciences institutions (University of Minnesota Rochester, Saint Camillus), and a comprehensive research university (University of Antwerp). A systematic audit of disambiguation artifacts is beyond the scope of this paper, and the top of this table should be read with appropriate caution.

Table 10 restricts to the 629 institutions with at least 5,000 authors. The University of Antwerp leads outright ($\varepsilon_2 = 2.25$, $h_2 = 106$, h_2 rank 3 among all $\geq$100-author institutions), placing simultaneously in the top 5 on both raw h_2 and size-adjusted index within this subset. The Weizmann Institute of Science ($\varepsilon_2 = 1.74$), Caltech ($\varepsilon_2 = 1.69$, size-adjusted rank 3), Princeton ($\varepsilon_2 = 1.68$), and Erasmus University Rotterdam ($\varepsilon_2 = 1.66$) follow, all achieving high size-adjusted index relative to their

Table 9 Top 10 institutions by size-adjusted index $\varepsilon_2 = h_2/(c_2\, S^{1/\beta_1})$ ($c_2 = 2.422$, $\beta_1 = 2.991$) among those with at least 100 authors ($n = 8{,}118$).

Institution	ε_2	h_2	Authors	h_2 rank
University of Minnesota Rochester	3.03	52	348	503
Kettering University	2.56	72	1,539	150
Académie Nationale de Médecine	2.46	34	182	1,287
Center for Astrophysics Harvard & Smithsonian	2.45	59	963	339
Saint Camillus International Univ. of Health Sci.	2.44	35	204	1,253
Université Bourgogne Franche-Comté	2.32	51	730	525
Stockholm Resilience Centre	2.26	29	146	1,806
University of Antwerp	2.25	106	7,207	3
Barcelona Institute of Science and Technology	2.25	74	2,462	125
Rush University	2.23	54	980	451

size. Among the elite comprehensive research universities that anchor the raw h_2 top 10, Caltech (size-adjusted rank 3 in this subset), the University of Geneva (rank 9), and Université Paris-Saclay (rank 69) each achieve $h_2 \geq 93$ with under 25,000 authors.

Table 10 Top 10 institutions by size-adjusted index $\varepsilon_2 = h_2/(c_2\, S^{1/\beta_1})$ ($c_2 = 2.422$, $\beta_1 = 2.991$) among those with at least 5,000 authors ($n = 629$).

Institution	ε_2	h_2	Authors	h_2 rank
University of Antwerp	2.25	106	7,207	3
Weizmann Institute of Science	1.74	76	5,711	117
California Institute of Technology	1.69	93	11,348	26
Princeton University	1.68	87	9,591	45
Erasmus University Rotterdam	1.66	81	7,943	79
The University of Adelaide	1.64	91	11,671	34
University of Colorado Anschutz Medical Campus	1.59	67	5,146	208
Vrije Universiteit Amsterdam	1.59	80	8,757	92
University of Geneva	1.58	80	8,847	91
City University of Hong Kong	1.57	70	6,062	167

External validation against Scimago Institutions Rankings

Joining the Scopus-derived Scimago Institutions Rankings (SIR) 2026 Overall Rank export to Table 9 yields 3,571 matched institutions: 33.0% of SIR's list and 44.0% of the paper's size-adjusted table.

On the matched set, the Spearman rank correlation between raw h_2 and SIR's Global Rank is $\rho = 0.798$ ($p \approx 0$); between ε_2 and SIR's Global Rank it is $\rho = 0.443$ ($p \approx 10^{-191}$). Restricting the match to SIR's "Universities" sector only (5,491 of SIR's 10,827 institutions, excluding ministries, hospitals, and companies that SIR also ranks) gives essentially the same result ($\rho = 0.802$ raw, $\rho = 0.441$ size-adjusted, $n = 3{,}515$), so the gap between the two correlations is not an artefact of comparing an academic h_2 ranking against a mixed-sector SIR list.

Against SIR's composite Global Rank, a source with no shared data infrastructure with OpenAlex, raw h_2 tracks SIR's Global Rank substantially more closely than ε_2 does. The lower ε_2–SIR correlation neither validates nor invalidates the size adjustment: SIR's Global Rank is a composite that rewards institutional scale and visibility, factors that raw h_2 partially captures and that dividing out size removes by construction. I report both comparisons as description of what each metric measures relative to SIR's composite.

3.10 h_3: country level

Aggregating institutional h_2 up to the country level via the same rank-based procedure yields h_3. Table 11 lists the ten highest-h_3 countries, and Figure 2 maps h_3 for all 214 countries with at least one qualifying institution.

Table 11 Top 10 countries by h_3.

Country	h_3	Institutions
United States	68	3,652
China	53	1,071
Germany	47	469
United Kingdom	46	472
Italy	44	192
Japan	44	1,145
France	40	374
Spain	38	140
Australia	36	115
South Korea	35	332

The United States leads by a wide margin, but Italy's 44 on only 192 institutions and Japan's 44 on 1,145 are a first hint that raw h_3 rewards national research-system size as well as depth.

3.11 Country size-adjusted index

Figure 3 repeats the size-scaling analysis at the country level, plotting h_3 against R^{1/β_2} for each country. The no-intercept fit gives slope $c_3 = 3.534$ and uncentered $R^2 = 0.900$ (centered $R^2 = 0.753$): institution count accounts for the large majority of variation in h_3 across countries. An OLS log-log regression of $\log(h_3)$ on $\log(R)$ gives slope $b = 0.502$ (implied $\beta \approx 1.99$, centered $R^2 = 0.795$), versus Egghe's $\beta_2 = 2.847$ (implied slope 0.351). As at the institution level, Egghe's exponent underestimates the empirical size-scaling; the normalization partially but not fully removes size dependence. A regression of $\log(\varepsilon_3)$ on $\log(R)$ across countries with ≥ 5 institutions ($n = 150$) yields slope $= 0.151$, $R^2 = 0.260$. Country-level size-adjusted rankings should therefore be interpreted with particular caution.

Table 12 ranks the 150 countries with at least 5 qualifying institutions by size-adjusted index $\varepsilon_3 = h_3/(c_3 R^{1/\beta_2})$. The Italy–Japan contrast foreshadowed in

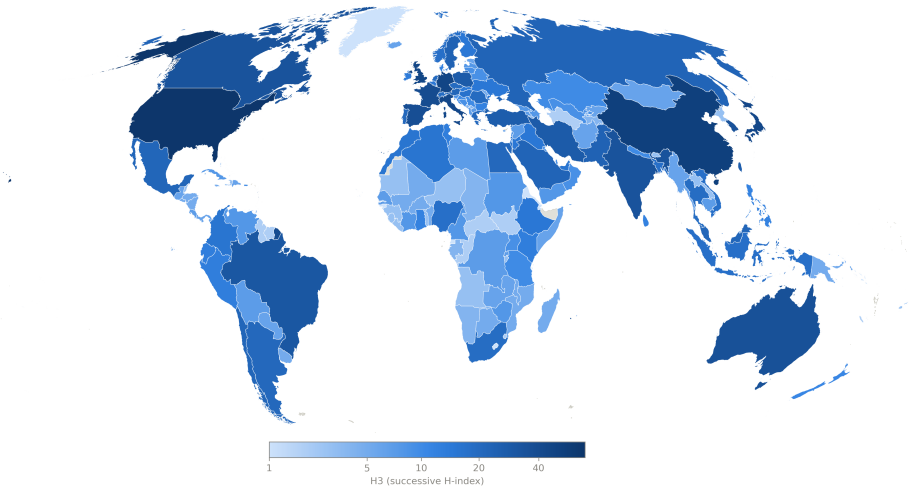


Fig. 2 h_3 by country. Countries with no qualifying institution are shown in grey.

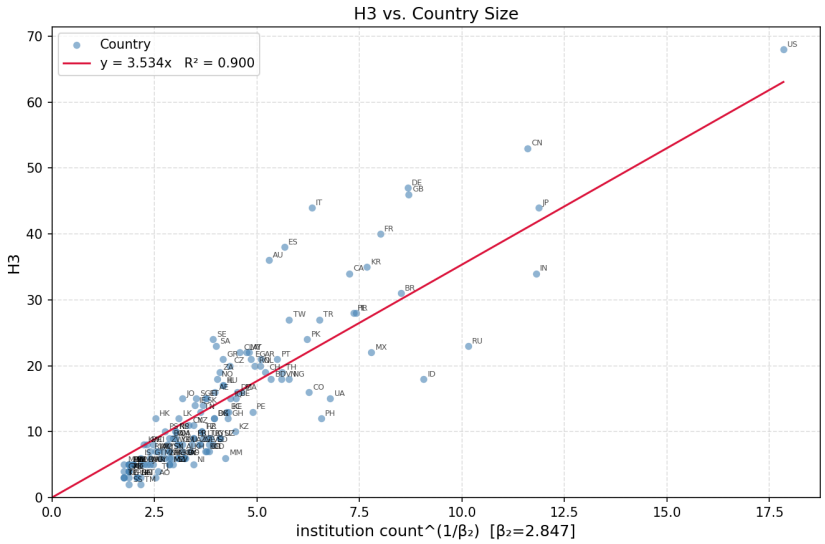


Fig. 3 Country-level h_3 versus R^{1/β_2} (Egghe's predicted scaling, $\beta_2 = 2.847$). No-intercept regression: $h_3 = 3.534\, R^{1/\beta_2}$, uncentered $R^2 = 0.900$ (centered $R^2 = 0.753$).

Subsection 3.10 is quantified: Italy ($\varepsilon_3 = 1.96$, rank 1) achieves $h_3 = 44$ from only 192 institutions, while Japan (size-adjusted rank 41, $\varepsilon_3 = 1.05$) reaches $h_3 = 44$ from 1,145 institutions, the same headline number but at 6× the institutional scale. Australia (rank 2, $\varepsilon_3 = 1.92$, 115 institutions) and Spain (rank 3, $\varepsilon_3 = 1.90$, 140 institutions) round out a top tier of countries with compact research systems that place well on this metric. Italy, Australia, Spain, Germany (rank 6), the United Kingdom (rank 7),

and France (rank 9) are the six countries that place in the top 10 on both raw h_3 and the size-adjusted index. Saudi Arabia is a notable outlier at 5th by size-adjusted index (h_3=23 on just 52 institutions, $\varepsilon_3 = 1.62$) despite ranking only 21st by raw h_3; prior reporting has documented practices in which high-h-index researchers list Saudi institutions as secondary affiliations in exchange for funding (Bhattacharjee, 2011), but this dataset cannot independently confirm that mechanism.

The United States falls to size-adjusted rank 37 ($\varepsilon_3 = 1.08$) despite its raw lead ($h_3 = 68$, rank 1). With 3,652 qualifying institutions, about 3.4× more than China and 19× more than Italy, even a large absolute h_3 advantage is absorbed by the size denominator. China (rank 19, $\varepsilon_3 = 1.29$, 1,071 institutions) is more compact and appears above the US on this metric.

Table 12 Top 10 countries by size-adjusted index $\varepsilon_3 = h_3/(c_3\, R^{1/\beta_2})$ ($c_3 = 3.534$, $\beta_2 = 2.847$) among those with at least 5 institutions ($n = 150$).

Country	ε_3	h_3	Institutions	h_3 rank
Italy	1.96	44	192	5
Australia	1.92	36	115	9
Spain	1.90	38	140	8
Sweden	1.73	24	49	19
Saudi Arabia	1.62	23	52	21
Germany	1.53	47	469	3
United Kingdom	1.50	46	472	4
Greece	1.43	21	58	28
France	1.41	40	374	7
Chile	1.36	22	76	23

3.12 Sensitivity to disambiguation error

Table 13 reports, for seven simulated population-wide error rates spanning $p = 1\%$ to a deliberately implausible $p = 80\%$ (Subsection 2.9), the Spearman rank correlation and top-set overlap between the perturbed and unperturbed rankings, averaged over 50 trials, together with how often the paper's three headline field-leadership claims survive the perturbation intact. The cross-validation sample's own raw outlier fraction ($4/30 \approx 13\%$ across both waves, Subsection 3.1) is included as an extra row for reference, though it carries no special status in this analysis.

Institution- and country-level rank correlations decline gradually with p: Spearman $\rho(h_2)$ falls from 0.9995 at $p = 1\%$ to 0.9793 at $p = 80\%$; $\rho(h_3)$ from 0.9998 to 0.9946. This follows from how h_2 and h_3 are computed: each is an order statistic, so what matters for a given institution or country is whether a noisy author sits near its own threshold, not how many noisy authors exist elsewhere; and because the perturbation is applied simultaneously to all institutions, individual rank shifts are dampened at the aggregate level. Top-20 institution overlap follows a U-shaped pattern: 95% at $p = 1\%$, declining to a minimum of 44% near $p = 20\%$, then recovering to 83% at

$p = 80\%$. This reflects a cancellation effect: at very high error rates, roughly equal numbers of inflation and deflation events accumulate within every institution's author pool, partially restoring the pre-perturbation rank ordering among the top-ranked set.

The field-specific leadership claims differ sharply in sensitivity. UCAS's Environmental Science lead survives all 50 trials at every error rate tested. Carnegie Mellon's Computer Science lead is robust at $p \leq 5\%$ (48/50 trials) but displaced at $p = 10\%$ (6/50) and $p = 13\%$ (1/50), recovering partially at $p = 80\%$ (16/50) as the cancellation effect sets in. Wageningen's Agricultural and Biological Sciences lead is the most sensitive: it holds 50/50 at $p = 1\%$ but falls to 36/50 at $p = 5\%$, and to 0/50 at $p = 10\%$ through $p = 40\%$, with partial recovery at $p = 80\%$ (1/50). Wageningen's greater sensitivity relative to CMU is consistent with the country-stratified error model: several of Wageningen's main competitors in Agricultural and Biological Sciences are Chinese institutions, which receive 3× the base error rate and therefore a larger expected perturbation.

The high Spearman correlation coexisting with low top-set retention ($p = 13\%$, $\rho(\mathrm{h}_2) = 0.9946$ but only 47% of the top-20 institutions are retained) is not contradictory: the Spearman correlation is dominated by thousands of institutions far from the top-20, where perturbations cause small rank shifts. The top-20 is populated by institutions often separated by only 1–3 h_2 points, making headline rankings sensitive to exactly the threshold-straddling errors the Monte Carlo applies.

Institution and country rank correlations remain high across all tested rates ($\rho(\mathrm{h}_2) \geq 0.979$; $\rho(\mathrm{h}_3) \geq 0.994$), and the UCAS Environmental Science claim is similarly stable. The Wageningen Agricultural and Biological Sciences and Carnegie Mellon Computer Science field-leadership claims are sensitive above $p \approx 5\%$ and should be read with correspondingly greater caution as single-institution results. The overall finding is that h_2 and h_3 rankings are structurally determined at population scale. Individual author errors are absorbed across many threshold comparisons, and at very high error rates the bidirectional perturbations increasingly cancel within institutions.

The three additional scenarios from Subsection 2.9 test the robustness of this picture. Under *directional inflation-only*, the absence of cancellation makes moderate error rates more destructive: top-20 institution overlap falls to 55% at $p = 5\%$ (versus 77% bidirectional) and Wageningen's Ag&Bio lead is displaced in all 50 trials at $p = 5\%$ (versus 37/50 surviving bidirectional); Carnegie Mellon's Computer Science lead holds in only 2 of 50 trials at $p = 5\%$ (versus 47/50). At $p = 13\%$, directional and bidirectional converge in top-20 overlap (44% vs. 47%) because both are already near their minimum: directional lacks the bidirectional recovery at high p up to this point. Under *threshold-concentrated* perturbation, the top-20 overlap stays at or above 95% across all tested rates including $p = 80\%$, and all three field-leadership claims hold in 47–50 of 50 trials through $p = 40\%$. This high stability arises because threshold-straddling errors, while in principle the most potent per author, still apply simultaneously to all competing institutions and tend to cancel within each institution's pool under bidirectional perturbation; concentrated unidirectional errors at a *single* institution's threshold would be more damaging. Under the *uniform (non-stratified)* model, top-20 overlap at $p = 13\%$ rises to 87%, and Wageningen holds rank 1 in all 50 trials (CMU in 47/50), confirming that the 3× country multiplier applied

to CN/KR/IR/SA institutions is the primary driver of the bidirectional scenario's instability.

Table 13 Sensitivity of h_2 and h_3 rankings to simulated disambiguation error (50 trials per rate; bidirectional perturbation with country-stratified rates for CN/KR/IR/SA; see Subsection 2.9). Overlap is computed on the top 20 institutions (by h_2) and top 10 countries (by h_3). The last three columns give the fraction of trials in which the named institution keeps rank #1 in the stated field.

p	$\rho(h_2)$	Top-20	$\rho(h_3)$	Top-10	Wag. Ag&Bio #1	UCAS EnvSci #1	CMU CS #1
1%	0.9995	95%	0.9998	99%	50/50	50/50	50/50
5%	0.9977	78%	0.9992	99%	36/50	50/50	48/50
10%	0.9958	54%	0.9986	97%	0/50	50/50	6/50
13%*	0.9946	47%	0.9983	96%	0/50	50/50	1/50
20%	0.9922	44%	0.9977	92%	0/50	50/50	0/50
40%	0.9865	56%	0.9965	90%	0/50	50/50	0/50
80%	0.9793	83%	0.9946	90%	1/50	50/50	16/50

*cross-validation sample's raw outlier fraction, included for reference only.

4 Conclusion

Schubert closed his 2007 note wondering whether h_2 of institutes would prove more interesting than h_2 of journals, absent the researcher–institute–country data needed to check. Nineteen years and 13.3 million OpenAlex authors later, the answer is yes, but not in the way a single top-10 table would suggest.

The overall h_2 ranking is the least interesting object in this paper. Because Medicine alone accounts for 24% of the author pool, the unrestricted university-wide aggregate is substantially shaped by biomedical field representation: institutions without a large biomedical program are structurally disadvantaged regardless of their non-biomedical strength. The field decomposition reveals a completely different pattern of field-level h_2 rankings: Wageningen ranks first in Agricultural & Biological Sciences by field h_2 while ranking 148th overall; the University of Chinese Academy of Sciences (UCAS) ranks first in Environmental Science, Materials Science, and Engineering by field h_2 from 24th overall, though these leads hold only under the paper's single-field assignment: UCAS leads Environmental Science at all multi-field thresholds tested, but CMU's Computer Science lead falls to Stanford under the more aggressive 20% threshold (Subsection 3.4); Carnegie Mellon ranks first in Computer Science by field h_2 from 159th under the single-field assignment; the University of Science and Technology of China ranks first in Energy from 66th; and Shenyang Pharmaceutical University ranks first in Pharmacology from 957th. Brazil offers perhaps the sharpest illustration of what the overall ranking hides: the Universidade de São Paulo and the Universidade Estadual de Campinas rank among the top six globally in Dentistry by field h_2 despite neither country nor either institution appearing anywhere near the top of any general-purpose ranking. The field-level h_2 is the unit that should be the default for comparison, not the aggregate.

On the methodological side, Egghe's power-law functional form is empirically well-supported: the no-intercept regression of h_2 on S^{1/β_1} gives uncentered $R^2 = 0.909$

(centered $R^2 = 0.810$), and the analogous regression for h_3 gives uncentered $R^2 = 0.900$ (centered $R^2 = 0.753$); size accounts for the large majority of variation at both levels. However, a residual size association remains after normalization (institution level: slope 0.073, $R^2 = 0.062$; country level: slope 0.151, $R^2 = 0.260$), so the size-adjusted index substantially but does not fully remove size dependence. However, the theoretical compounding prediction $\alpha_0\alpha_1 \approx \beta_1$ fails by 146%. The model is empirically useful; the underlying theory needs a different reading.

Against Scimago Institutions Rankings 2026 (SCImago Research Group, 2026), a Scopus-derived ranking independent of OpenAlex, raw h_2 correlates at $\rho = 0.798$ against SIR's Global Rank versus $\rho = 0.443$ for ε_2, on a smaller, name-matched set of 3,571 institutions (Subsections 2.12 and 3.9). Against SIR's composite Global Rank, raw h_2 tracks the external ranking substantially more closely than ε_2 does, plausibly because SIR's composite is itself influenced by scale-correlated factors that dividing out size removes; the lower ε_2–SIR correlation neither validates nor invalidates the size adjustment; it is a description of the metrics' behaviour relative to a composite that itself rewards scale.

Several limitations constrain interpretation. Author disambiguation is the most consequential open problem: the four outlier cases identified in cross-validation (notably Lei Jiang, $h_1^{\text{OpenAlex}} = 215$ against Web of Science = 115) confirm that merger errors cluster in non-Anglophone and common-name strata, exactly the authors whose inflated h_1 values would propagate most damagingly into institutional and country-level rankings. That cross-validation sample was deliberately adversarial, however, and cannot by itself say how much of the full 13.3-million-author population is affected; a Monte Carlo sensitivity analysis run at population-wide error rates from 1% to a deliberately implausible 80% (Subsections 2.9, 3.12) finds institution- and country-level rank correlations declining gradually but remaining high across all tested rates ($\rho(h_2) \geq 0.979$; $\rho(h_3) \geq 0.994$). However, top-20 institutional retention falls to 47% at the cross-validation sample's raw 13% outlier rate; a high rank correlation across thousands of institutions can coexist with substantial headline-ranking churn because the correlation is dominated by the bulk of the distribution, while the top-20 is populated by institutions often separated by only 1–3 h_2 points. The UCAS Environmental Science claim is stable throughout, while the Wageningen Agricultural and Biological Sciences and Carnegie Mellon Computer Science field-leadership claims are sensitive above $p \approx 5\%$ and should be read with correspondingly greater caution. Arts and Humanities are systematically under-covered in all major citation databases: both threshold authors sampled at Oxford returned no value in Web of Science, Scopus, or Google Scholar, and h-index comparisons in book-based fields should be read with correspondingly less confidence. Field-specific rankings are also sensitive to how field membership is defined: a multi-field assignment (authors counted in every field comprising $\geq 33\%$ of their output) yields highly stable overall rankings ($\rho = 0.960$); UCAS retains its Environmental Science lead at all multi-field thresholds tested, while Carnegie Mellon's Computer Science lead falls to Stanford at the more aggressive 20% threshold (Subsection 3.4); these findings confirm that the field-leadership claims should be read as reflecting depth of specialisation rather than unconditional supremacy. OpenAlex's inconsistent retraction flagging leaves a small, unquantified

fraction of h_1 values potentially inflated. Finally, h_2 is not comparable across fields of different size: Decision Sciences (max $h_2 = 20$) cannot be placed on the same scale as Medicine (max $h_2 = 118$), and the results should be read within fields only.

A direct numerical comparison between Schubert's successive-index hierarchy and the k-step h-index of Yang et al. (2023) on a common institutional sample would clarify how the two complementary constructions differ in practice. Reimplementing the algorithm of Yang et al. over OpenAlex's full citation graph at institutional scale is beyond the scope of this paper and remains a natural direction for future work.

The most impactful near-term improvement to this framework would require a data source that does not yet exist: departmental affiliation in OpenAlex. If OpenAlex were to record the department within an institution alongside the institution itself, the hierarchy would extend naturally to researcher → department → institution → country, with successive indices h_1 through h_4. Departments are field-specialized by construction, so the step of inferring each author's modal field from topic weights may become unnecessary at the departmental level. I flag this as the single most valuable addition OpenAlex could make for scientometric work of this kind.

Statements and Declarations

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